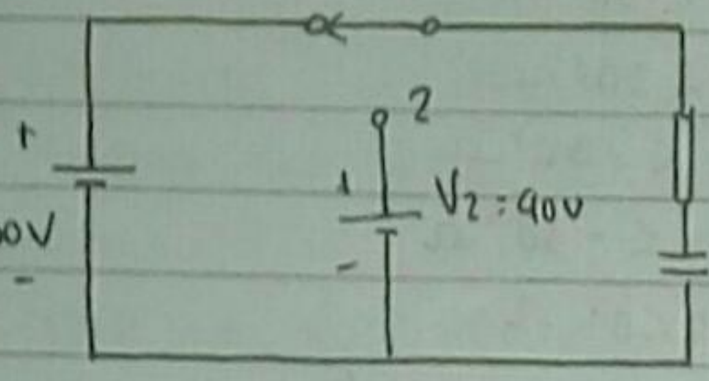
Nama : Khoirul Tasya
NIM : 19133109 99
Kelas : 1B TTG

بسم الله الرحمن الرحيم

15 - 7 - 20

UAS RL 2

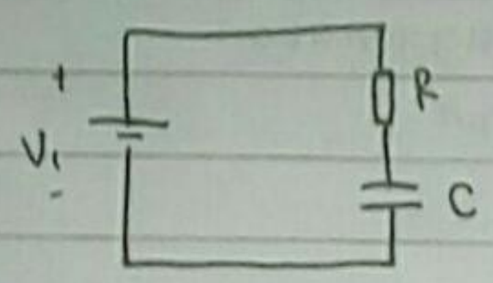
1.) Dik rangkaian DC



Hitung dan gambarkan pada kertas grafik gelombang $i(t)$; $V_R(t)$ dan $V_C(t)$ kerika saklar di posisi 1 dan 2 bila saklar pindah ke posisi 2 saat $t = 50\ mS$, $V_0 = 0$

Jawab

Saklar 1



$$V_C = V_S\left(1 - e^{-\frac{1}{RC}t}\right)$$

$$= V_1\left(1 - e^{-\frac{1}{RC}t}\right)$$

$$= 100\left(1 - e^{-100t}\right)\ V$$

$$V_{R(t)} = V_S - V_C$$

$$= 100 - 100\left(1 - e^{-100t}\right)$$

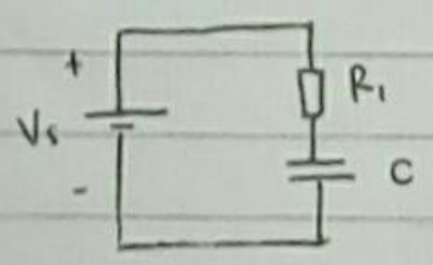
$$= 100 - 100 + 100e^{-100t}$$

$$= 100e^{-100t}\ V$$

$$I_1(t) = \frac{V_S}{R_1} \cdot e^{-\frac{1}{RC}t}$$

$$= \frac{100}{10} e^{-100t} = 10e^{-100t}\ A$$

Saklar 2



pada saat pindah ke saklar 2 $t = 50ms$

$$V_C(t = 50ms) = 100\left(1 - e^{-100 \cdot 50 \cdot (10^{-3})}\right)$$

$$= 100\ Volt$$

$$V_C = V_0' = 100\ Volt$$

$$V_C = V_S - (V_S - V_0')e^{-100t}$$

$$= 40 - (40 - 100)e^{-100t}$$

$$= 40 - (-60e^{-100t})$$

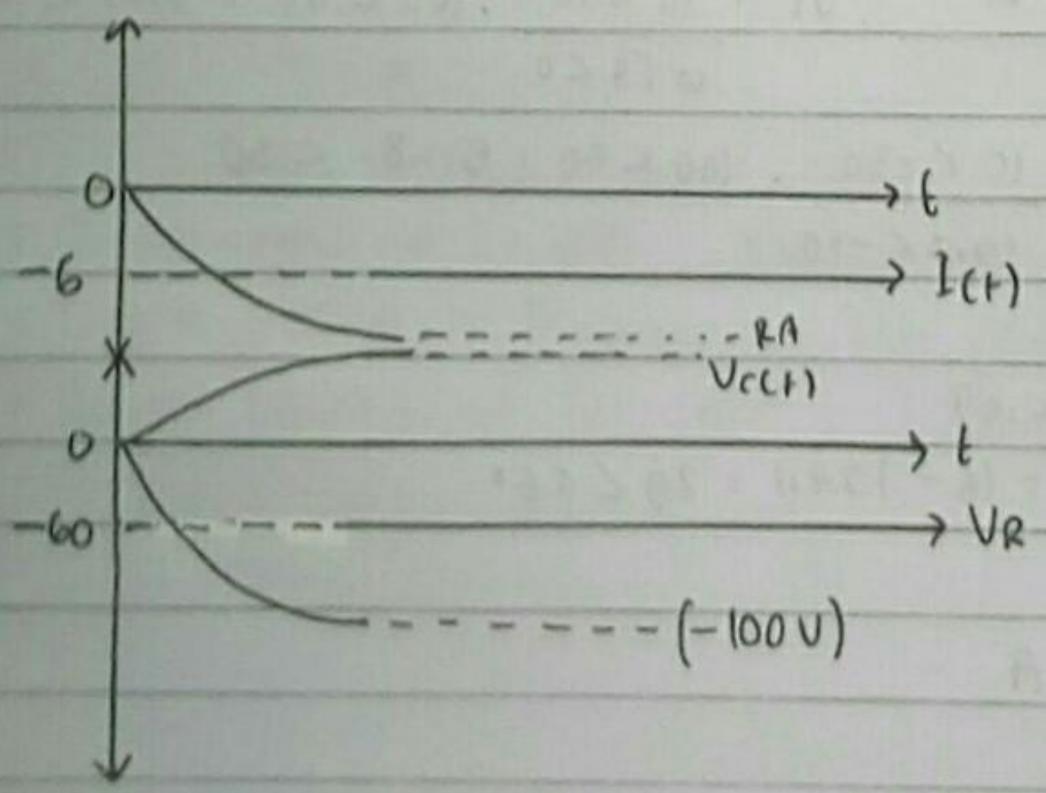
$$= 40 + 60e^{-100t}\ V$$

$$V_R = -V_C(t)$$

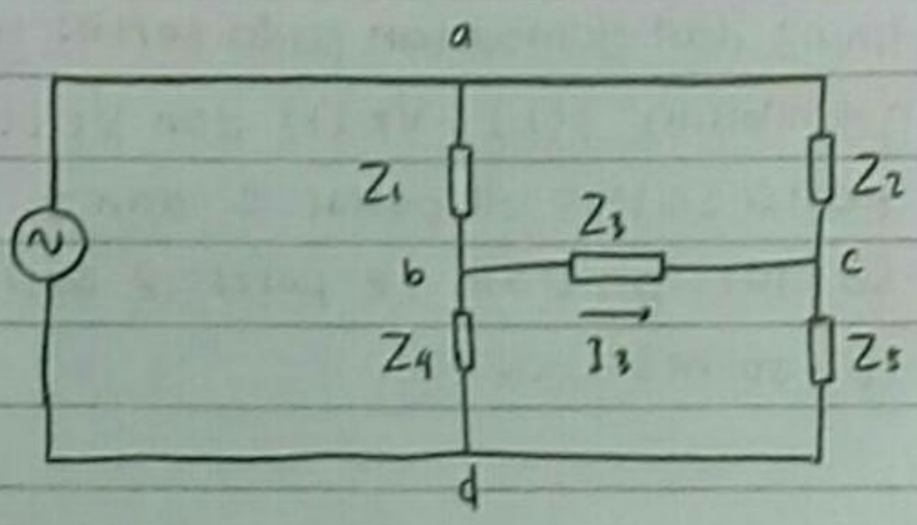
$$= -(40 + 60e^{-100t})$$

$$= -40 - 60e^{-100t}\ V$$

$$I_2(t) = \frac{V_R}{R} = \frac{-40 - 60e^{-100t}}{10} = -4 - 6e^{-100t}\ A$$



2) Diketahui Rangkaian RL



$V_S = 100 \angle 30° V$

$Z_1 = 10 \angle 30° \Omega$

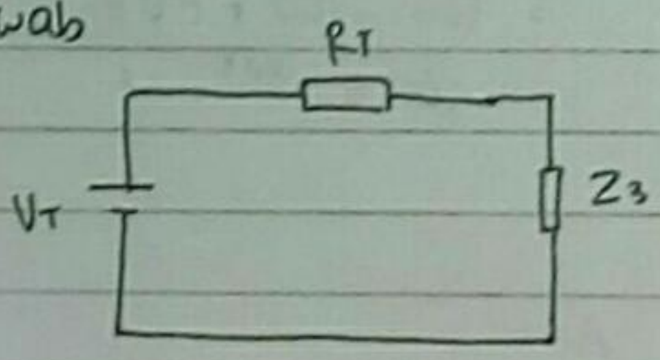
$Z_2 = 10 \angle -30° \Omega$

$Z_4 = 10 \angle -30° \Omega$

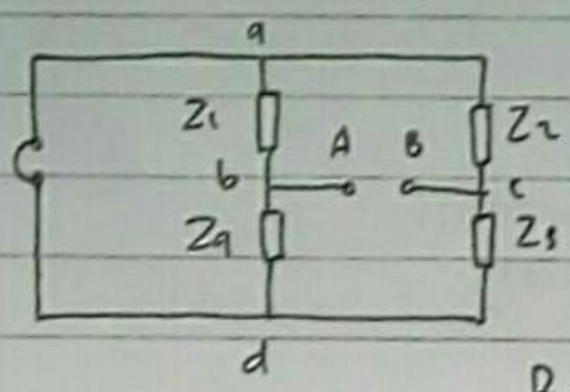
$Z_5 = 10 \angle 0° \Omega$

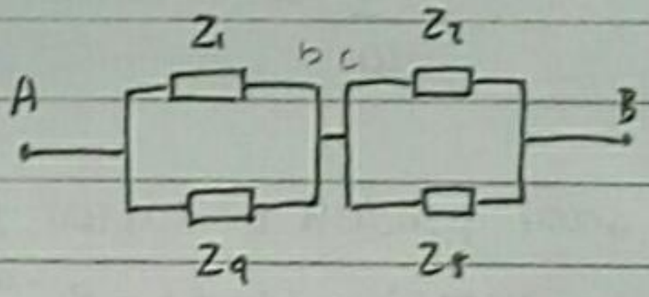
Tentukan Z_3 agar terjadi daya maksimum dan hitung I_3 dan daya maksimum. Boleh menggunakan Thevenin atau Norton

Jawab



Daya max RT = Z_3





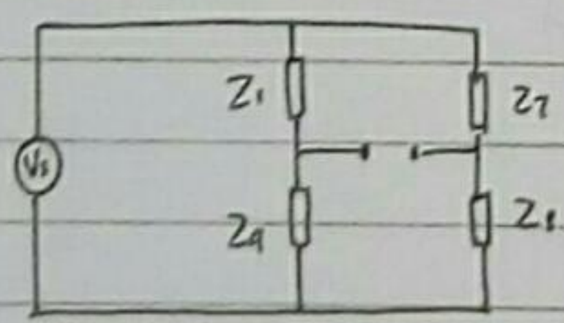
$Z_1 // Z_4 + Z_2 // Z_5$

$$R_{th} = \frac{10\angle 30 \cdot 10\angle -30°}{5\sqrt{3} + j5 + 5\sqrt{3} - j5} + \frac{10\angle -30 \cdot 10\angle 0}{5\sqrt{3} - j5 + 10 + j0}$$

$$= \frac{100\angle 0}{10\sqrt{3}\angle 0} + \frac{100\angle -30}{19.3\angle -30}$$

$R_{th} = 5.8\angle 0 + 5.2\angle 0 = 11\angle 0 \Omega$

$Z_3 = 11\angle 0$



$V_{AB} = V_{TH}$

$$V_{Z1} = \frac{Z_1}{\quad} \cdot V_S = \frac{10\angle 30}{10\sqrt{3}\angle 0} \cdot 100\angle 30 = 57.7\angle 60$$

$$V_{Z2} = \frac{10\angle -30}{19.3\angle -30} \cdot 100\angle 30 = 51.8\angle 30$$

$V_{TH} = V_A - V_B$

$= V_{Z1} - V_{Z2} = 57.7\angle 60 - 51.8\angle 30$

$= 28.85 + j50 - (44.85 + j25.9)$

$= -16 + j24.1$ (di balik) $= 16 - j24.1 = 29\angle 56°$

Daya max pada $Z_3 = V_{TH} \cdot I$

$$I = \frac{V_{TH}}{23 + R_{TH}} \cdot \frac{29\angle 56}{2.11\angle 0} = 1.32\angle 56 \text{ A}$$

Daya $= 29\angle 56 \cdot 1.32\angle -56 = 38.28\angle 0$

3) Sebuah generator AC 3 fasa 220 V/ 380V, mencatu beban 3 fasa sembang yang diwakili 3 buah impedansi : $Z = 22 \angle 30^\circ \Omega$

Hitung arus setiap fasa, arus setiap line dan dayanya untuk beban hubungan Bintang dan Delta

Jawab :

Delta → $V_{LL} = 220$ V

- $I_f = \frac{220 \angle 0^\circ}{22 \angle 30} = 10 \angle -30^\circ A \rightarrow \varphi = 30^\circ$

$I_{AB} = 10 \angle -30^\circ A$

$I_{BC} = 10 \angle -120 - 30 = 10 \angle -150^\circ A$

$I_{CA} = 10 \angle 120 - 30 = 10 \angle 90$

- $I_L = \sqrt{3} I_f = 10\sqrt{3}$

$I_a = I_L \angle -\varphi - 30^\circ = 10\sqrt{3} \angle -60^\circ$

$I_b = I_L \angle -120 - \varphi - 30 = 10\sqrt{} \angle -180^\circ A$

$I_c = I_L \angle 120 - \varphi - 30 = 10\sqrt{} \angle 60^\circ A$

- Daya semu → $S_{3\,fasa} = 3 V_f I_f \angle \varphi = 3 . 220 . 10 \angle 30^\circ = 6600 \angle 30^\circ$ VA
- Daya nyata → $P_{3\,fasa} = 3 V_f I_f \cos\varphi = 3 . 220 . 10 \cos 30^\circ = 5715,76$ watt
- Daya reaktif → $Q_{3\,fasa} = 3 V_f . I_f \sin\varphi = 3 . 220 . 10 \sin 30^\circ = 330$ VAR, lag

Bintang → $V_{LL} = 380$ V

- $\dot{I}_f = \frac{380 \angle 0^\circ}{22 \angle 30^\circ} = \frac{190}{11} \angle -30^\circ A$

$I_a = \frac{190}{11} \angle -30^\circ A$

$I_b = \frac{190}{11} \angle -150^\circ A$

$I_c = \frac{190}{11} \angle 90^\circ A$

$I_b = I_f = \frac{190}{11} A$

- Daya semu → $S_{3\,fasa} = 3 V_f I_f \angle \varphi = 19690,9 \angle 30^\circ$ VA
- Daya nyata → $P_{3\,fasa} = 3 V_f . I_f \cos\varphi = 17052,8$ watt
- Daya reaktif → $Q_{3\,fasa} = 3 V_f I_f \sin\varphi = 9895,95$ VAR lag

4) Sumber satu fasa 220 volt mencatu beban sebagai berikut:

- beban 1 : (220 volt ; 4A ; 0,5 lag)
- beban 2 : (220 volt ; 110 VA ; 55$\sqrt{2}$ watt ; 55$\sqrt{2}$ var lag
- beban 3 : (220 volt ; 55 watt ; 55$\sqrt{3}$ var lag)
- beban 4 : (220 volt ; (132 - j176)Ω

Hitung daya total (S . P . Q) . arus total dan faktor daya total

Bila diinginkan faktor daya total adalah 0.9 lag, hitung kapasitansi kapasitor yang harus dipasang

Jawaban

* Beban 1 → $S_1 = V.I$: $880 \angle 60$ VA ; $P_1 = S\cos\theta$: 440 watt ; $Q_1 = S\sin\theta$: $440\sqrt{3}$ Var

* Beban 2 → $S = V.I$; $I_2 = \frac{S}{V} = \frac{110}{220} = 0{,}5$ A ; $S_2 = 110 \angle 45°$ VA ; $P_2 = 55\sqrt{2}$ watt ; $Q_2 = 55\sqrt{2}$ Var

* Beban 3 → $S_3 = \sqrt{P_3^2 + Q_3^2}$; $S_3 = 110 \angle 60$ VA ; $P_3 = 55$ watt ; $Q_3 = 55\sqrt{3}$ Var

* Beban 4 → $Z_4 = \sqrt{(132)^2 + (176)^2} = \sqrt{98400} = 220$; $Z_4 = 220 \angle -53\ \Omega$; $I_4 = \frac{V}{Z} = 1 \angle 53$ A ; $S_4 = V.I = 220 \angle -53$ VA ; $P_4 = 132$ watt ; $Q = 176$ VAR

P total : $(440 + 55\sqrt{2} + 55 + 132)$ watt = 705 watt

Q total : $440\sqrt{3} + 55\sqrt{2} + 55\sqrt{3} + (-176)$ = 759 watt

S total : $705 + j759$ VA = $1036 \angle 47°$ VA

Jika diinginkan $\cos\varphi' = 0.9$ lag

$P_{to}' = P_{to} = 705$ watt

$|S_{to}'| = \frac{P_{to}'}{\cos\varphi'_{to}} = \frac{705}{0{,}9} = 783$ VA

$Q_{to}' = \sqrt{S_{to}'^2 - P_{to}'^2} = 391$ Var

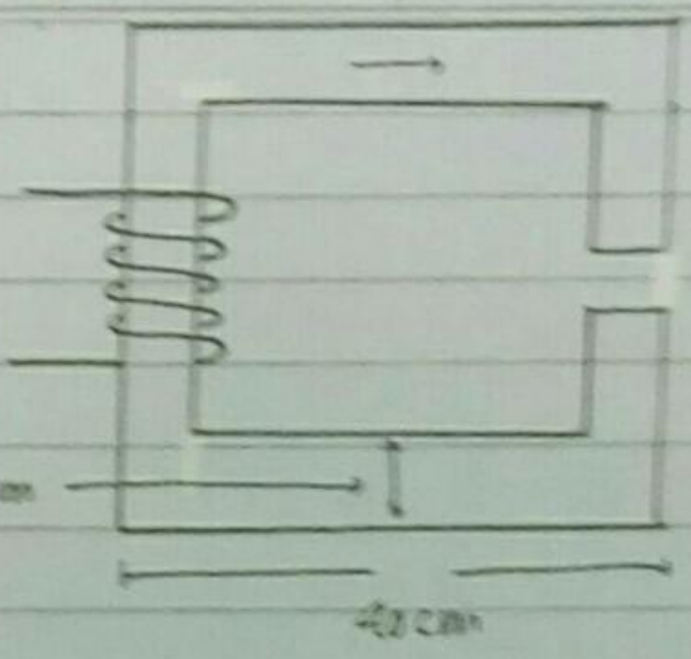
$\Delta Q = Q_{to} - Q_{to}' = 378$ Var

$Q_c = \Delta Q = 378$ Var

Nilai kapasitor yang harus dipasang

$C = \frac{Q_c}{\omega V^2} = 25\ \mu F$

5) Tentukan nilai fluks magnit yang mengalir dan nilai reluktansinya. bila inggi dan lebar Sama yaitu 40 cm. tebal 5cm dan lebar celah udara 5 cm.



Jawab :

Reluktansi $R_m = \frac{k}{\mu_0 A} = \frac{40+40+40\,(40-5)}{4\times10^{-7}\times5\times5}$

$= \frac{155}{100\times10^{-7}} = \frac{155}{10^{-5}} = 155\times10^{5}$ A/wb

fluks : $\phi = \mu_0 A = 4\times10^{-7}\times5\times5\times10^{-5} = \frac{1}{10}$ W